MACHINE LEARNING

Logistic Regression

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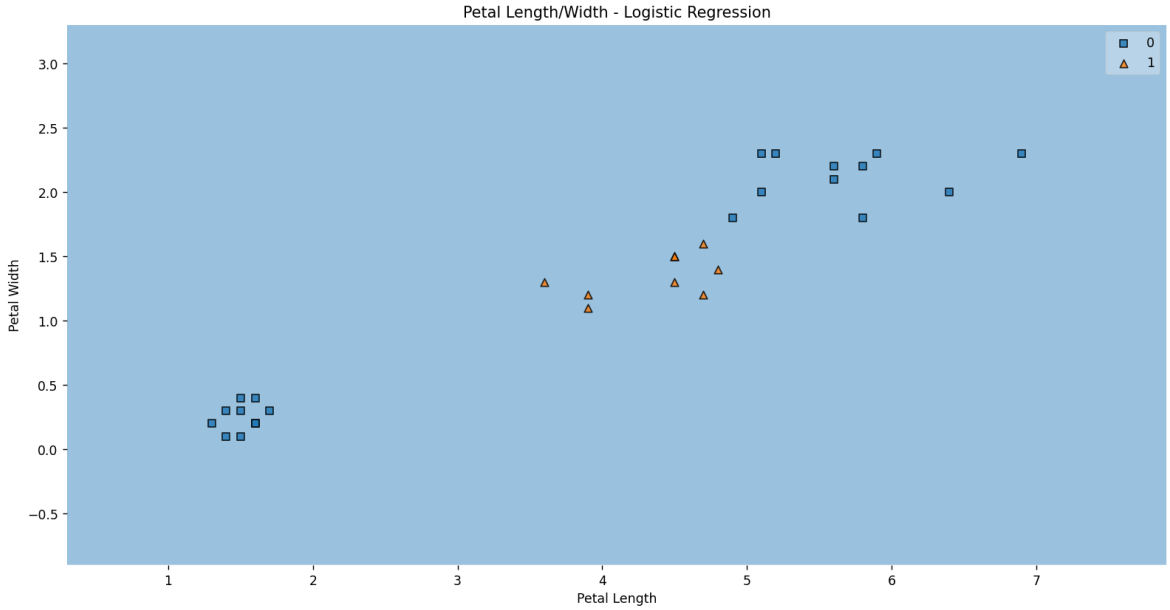
Logistic Regression

1.

Here is the graph of logistical regression for the file train_classifier1.py with petal length versus petal width.



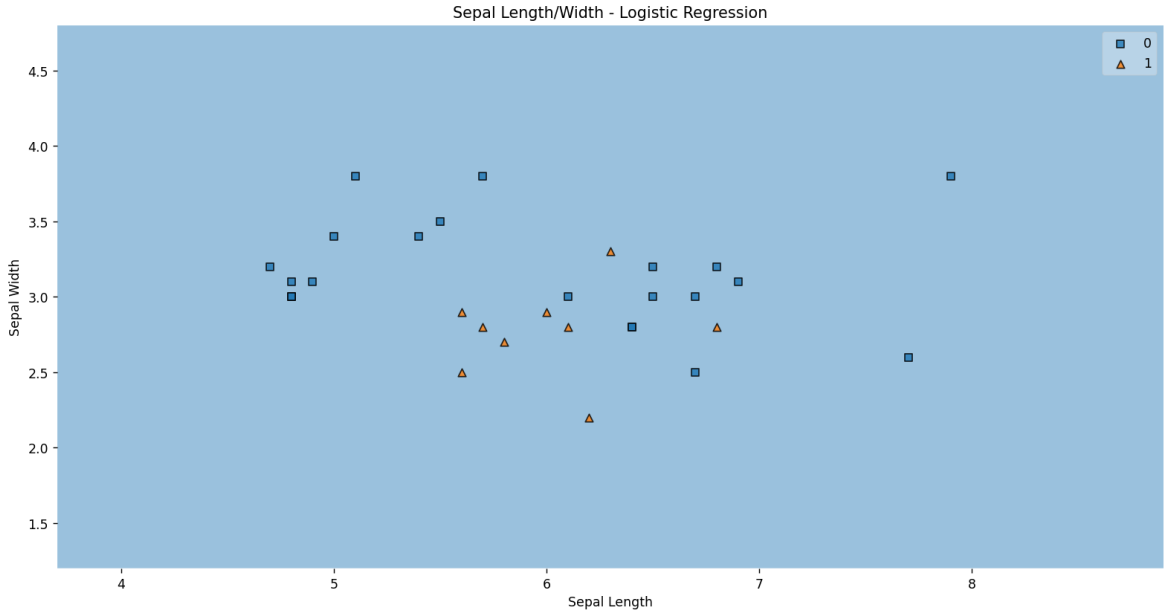

```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/train_classifier1.py"
Early stopping at epoch 10
```

Here is the accuracy output of eval_classifier1.py

```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/eval_classifier1.py"
Available keys: ['Weights', 'bias']
Accuracy is: 0.7
```

2.

Here is the graph of logistical regression for the file train_classifier2.py with sepal length versus sepal width




```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/train_classifier2.py"
Early stopping at epoch 5
```

Here is the accuracy output of eval_classifier2.py

```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/eval_classifier2.py"
Available keys: ['Weights', 'bias']
Accuracy is: 0.7
```

3.

Here is the graph of logistical regression for the file train_classifier3.py

We will not get any graph because we are using all the attributes.

```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/train_classifier3.py"
Early stopping at epoch 5
```

Here is the accuracy output of eval_classifier3.py

```
PS C:\Users\USER\OneDrive\Desktop\UTA\2nd SEM\Tharunproject> & C:/Users/USER/anaconda3/python.exe "c:/Users/
USER/OneDrive/Desktop/UTA/2nd SEM/Tharunproject/eval_classifier3.py"
Available keys: ['Weights', 'bias']
Accuracy is: 0.7
```

Logistic Regression

1. Why is the softmax function used in multi-class logistic regression (Hint: the model itself produces logits)?

It is used to handle classification tasks with more than two classes. Converts logits into a probability distribution over multiple classes, making it suitable for multi-class classification problems